CPSY 1291
Computational Methods for Mind, Brain, and Behavior

Fall 2026

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Course Information

Instructor · Professor Thomas Serre

Email · thomas_serre@brown.edu

Office · Carney 419, 164 Angell Street

Office Hours · Wednesdays at 1:00 PM, Carney 419[1] — a short email beforehand is appreciated, so I know to expect you

Class Time · Tue/Thu, 2:30–3:50 PM

Location · Salomon Center 003

Recitations · Tuesdays, 6:30–7:50 PM, TA-led, optional (room on Canvas)

LMS · Canvas (assignments, grades, and course materials)

Communication · Ed Discussion

Teaching Assistants

Head TA · Peisen Zhou (PhD student, Cognitive & Psychological Sciences)

TAs · Lyfey Vutha · Tekin Gunasar

Staff office hours will be announced on Canvas before the semester begins.

Course Description

This course introduces *NeuroAI*, an interdisciplinary field at the intersection of brain science and artificial intelligence. You will develop a mathematical and computational toolkit for studying the brain, mind, and behavior, combining classical models with hands-on experience in modern machine learning.

The course is organized around three themes: describing representations (spaces, similarity and geometry, including how to read representations out and compare them); learn-

[1] If I am not in 419, I am usually either in the Innovation Hub, the open area you see first when you step out of the fourth-floor elevators, or on the other side of the floor, around the student open space.

ing representations, from the model neuron to deep networks, including vision, model–brain comparison and learning without labels; and dynamics, sequences and world models (recurrence, attractors, attention, language models and world models). Explainability runs throughout: we will repeatedly ask what a trained model computes, how it computes it, and whether the same account applies to biological systems. The full sequence of topics appears in the schedule.

Who Should Take This Course?

This advanced course is intended for students interested in the intersection of brain and cognitive sciences with AI. It draws on cognitive psychology, neuroscience, mathematics, and computer science; no single disciplinary background is assumed.

Prerequisites: Prior programming experience is required, but prior Python experience is not. You should be able to write and debug short programs, break problems into functions, and work with basic data structures. CPSY 0950 or 1292; an introductory CSCI course with a programming component (such as CSCI 0111, 0150, 0170, or 0190); APMA 0160; EEPS 0250; or equivalent experience meets this requirement. If your previous course used another language, attend the early recitations, which include Python bootcamp sessions. Linear algebra and statistics are recommended, not required.

Preparation for Research

This course is meant to prepare you to conduct research in NeuroAI. The assignments develop technical skill by asking you to implement the methods yourself. Lectures range more broadly and introduce the primary literature. The final project asks you to formulate and investigate your own question. Its research trail begins in week 2, giving you most of the semester to read in areas that may become your project.

Learning Objectives

By the end of this course, you will be able to:

1. **Implement core methods** from computational neuroscience and machine learning, including dimensionality reduction, Hebbian and error-driven learning rules, and deep neural networks.
2. **Analyze neural and behavioral data** with appropriate computational models and interpret the results in light of relevant theories.
3. **Compare artificial and biological systems**, identifying where an AI model succeeds or fails as a model of human cognition or the brain.

4. **Explain what a trained network computes** using attribution, feature visualization, and representational similarity analysis, and discuss mechanistic interpretations of attention.
5. **Assess models critically**, including their generalization, limitations, and failures as models of biological systems.
6. **Design and communicate a reproducible study** that brings together ideas from computer science, neuroscience, and cognitive psychology.

Course Activities

Plan for approximately **180 hours** over the semester: 33 hours in class; 72 hours on four programming assignments (about 18 hours each, on average); about 20 hours preparing for the two exams (roughly 10 hours per exam); and about 55 hours on the research trail, proposal, and final project. Optional recitations are additional to these 180 hours.

Assignments: Release and Due Dates

Assignments are released on a rolling schedule and remain open for about two weeks. Assignment 1 is posted on the first day of class and gets an extra week for students who are still shopping the course or coming up to speed in Python. Assignment 2 closes four days before Exam 1, and Assignment 3 opens on the day of the exam.

Joining the course late does not extend any deadline. Assignment 1's extra week is the allowance for shopping period and applies to everyone. Students who add the course after work has been released remain responsible for that work and for the published deadlines. If you are shopping this course, begin Assignment 1 before finalizing your registration.

	Released	Due	Covers	Weight
Assignment 1	Thu 9/10	Tue 9/29	L01–L04	10%
Assignment 2	Tue 9/29	Sun 10/18	L05–L09	10%
Assignment 3	Mon 10/26	Mon 11/9	L09–L13	10%
Assignment 4	Tue 11/10	Tue 11/24	L15–L17	10%

These assignments are designed to be worked on continuously. Assignments and lectures run in parallel: an assignment is posted before all of its supporting lectures have met. Start when the notebook is posted, complete the parts already covered in class, and return to the remaining parts as the material arrives. About 18 hours per assignment is an average planning estimate; actual time can be substantially lower or higher depending on programming experience.

Recitations

A weekly 80-minute recitation is led by the teaching assistants and is optional. Recitations include Python bootcamp sessions, review of the mathematics used in the course, hands-

on work with NumPy, scikit-learn, and PyTorch, lecture review, and assignment help. Attend the sessions you need. Recitations meet **Tuesdays, 6:30–7:50 PM**; the room will be posted on Canvas.

Exams

Both exams are taken **in class**: Exam 1 on **Thu 10/22** focuses on Lectures 1–8 and the four steps of backpropagation from Lecture 9, and Exam 2 on **Tue 12/1** focuses on Lectures 9–19. Exam 2 will not include standalone questions whose main subject is material from Lectures 1–8. Later topics build on earlier concepts and methods, so you may need the earlier material to answer questions about Lectures 9–19. Brown's official final-exam slot is used for the project poster session, not for Exam 2.

Exams are **written on paper**. Most questions ask you to explain an idea, work through a derivation, interpret a result, or defend a modeling choice; some ask you to explain or debug short code. This tests your ability to apply course skills, not memorize assignment solutions. I will provide sample questions beforehand. You may bring one sheet of notes, both sides, handwritten or typed. No textbooks, laptops, or phones are permitted.

Missed exams. If you miss either exam for an approved reason, we will arrange a **makeup exam**. Contact me as early as possible, before the exam whenever you can, and provide official documentation, which is required in every case (see *Extensions, Absences, and Late Work* below). Makeups are scheduled individually and as close to the original date as circumstances allow. If a makeup cannot be completed before grades are due, we will arrange an Incomplete under the relevant dean's procedure. Both exams remain required; their weights are not redistributed.

An exam missed without an approved excuse scores zero and, because all coursework is required, leaves the course requirements unmet. Tell me as early as possible about any known conflict, such as a religious observance, a scheduled medical procedure, or a University-sanctioned commitment.

Software, Computer Access, and Submission

Programming assignments are completed in Python in Jupyter notebooks that run on **Google Colab**. Sign in with your Brown account, not a personal one: course notebooks, datasets, and shared drives are released to Brown accounts and may be inaccessible from a personal account. Colab's free tier should be sufficient for the four programming assignments. The final project may require more compute; if GPU time becomes a bottleneck, we can provide access to Oscar, Brown's computing cluster. Setup instructions and compute details will be posted with the first assignment and covered in recitation.

Because this is a computational course, you will need regular access to a computer. You may use your own computer or one available through a campus computer lab or another shared facility. If your computer breaks or you temporarily lose access to one, Brown's **IT Service Center** offers short-term laptop loans on a first-come, first-served basis. Contact me as early as possible if reliable computer access is a concern.

Submit your notebooks through Canvas by **11:59 PM** on the due date, with all outputs already run. Grades and written feedback will be returned through Canvas.

Communication

Use Ed Discussion for questions about assignments, lectures, setup, readings, or the project. Post privately if your question would reveal part of a solution; otherwise post publicly so others can benefit. Reserve email for personal matters. I welcome course feedback on Ed, by email, or during office hours.

Set up Ed during the first week. Announcements about deadlines, assignment releases, and schedule changes are posted there. At the start of the semester:

- **Join the Ed course** using the link above or on Canvas. You will not receive Ed announcements until you join.
- **Set notifications to** *Instant* in your Ed profile settings. The default is not instant, so replies and new threads may otherwise reach you late or in a digest.
- **Add Ed's sender address to your contacts**, notifications@edstem.org, so its mail is not filtered into spam or promotions.

You are responsible for announcements posted to Ed. If you are unsure whether notifications are working, reply briefly to the welcome thread and confirm that the email reaches you.

Readings and Course Materials

The paper you choose for each research-trail entry is required; all other readings are optional. Slides link to primary sources for students who want to go deeper, and occasional supplementary readings may appear on Canvas. Slides will be available before class.

Cost of course materials: $0. There is no required textbook. All required readings are available at no cost through the Brown Library or as open-access papers, and the software used in the course (Google Colab, Python, and its libraries) is free.

I will try to record lectures and post them on Canvas. Recordings are a study aid, not a substitute for attending, and cannot be guaranteed.

Assessment

Assignments (4 programming assignments, 10% each)	40%
Project strand (research trail 7% + proposal 3% + project & poster 12%)	22%
Minute papers (in-class retrieval and feedback)	15%
Exam 1 (in-class; Thu 10/22)	9%
Exam 2 (in-class; Tue 12/1)	14%

Project Strand

The project strand begins in week 2 and is worth **22%**. All three components below are required:

- **Research trail** (7%) — four times, choose one citation from any lecture so far — usually a recent one, but reaching back is welcome, including to propose applying an earlier method to newer material — read the paper, and post four to six sentences on Ed: what it did; what surprised you or raised questions; and, from week 6, what you would test next. For each entry, reply once to a classmate. Deadlines are 11:59 PM on Fri 9/25, 10/9, 10/30, and 11/13; Canvas will announce any changes. The grade counts your **best three of the four** entries, so one entry may be skipped without penalty; the first comes after only a few lectures and is a good one to treat as practice.
- **Proposal** (3%) — your team submits **one** one-page plan by **Tue 11/24 at 11:59 PM**, allowing the teaching staff time to respond before the first project work session. The proposal may grow out of any team member's research-trail entry, but it need not; teams may propose any question within the scope of the course. Teams must be registered on Canvas by **Thu 11/19**.
- **Project, report, and poster** (12%) — work in teams of two (occasionally three, depending on enrollment). Your team presents a **poster** during the official exam slot, **Mon 12/21 at 9:00 AM**; the two-page report is due at the same time. The session is **open to the public**, and you may invite guests. Poster specifications and printing details will be posted on Canvas.

One submission per team, but individual grades. Each team submits a single proposal and a single report. Project grades are nonetheless individual, based on each student's contribution and demonstrated understanding.

Report format. Use the **two-page extended-abstract** format of Cognitive Computational Neuroscience (CCN). Begin with an abstract of no more than 300 words. Text, tables, and figures count toward the two-page limit; references do not. The **CCN Word template** and a PDF example are posted on Canvas, and teams are expected to use the template. The limit is strict: deciding what to include is part of the exercise. CCN uses this format for poster submissions.

Projects may extend a result, add a missing control, apply a method to a new dataset, or test a model's prediction on people. Carefully analyzed negative results are welcome; we grade the question, method, and interpretation. Class meetings from Thu 12/3 onward are project work sessions with the teaching staff.

Minute Papers and Attendance

When time permits, I reserve a few minutes at the end of class for a **minute paper**. The prompt varies, but I will usually ask you to identify three ideas that you found interesting, did not understand, or want to revisit — and to make at least one of them something you did not fully understand or are still wondering about; those are the most useful to me. Minute papers serve two purposes: they are how attendance is taken in this course, and — more importantly — the act of recalling the material and putting it into your own

words helps you remember it later.[2] They are not a quiz: credit is for checking in and making a genuine attempt, not for being right. Your responses also show me what the class understands and where I need to clarify. Whenever possible, I will address recurring questions in the next class or on Ed.

Minute papers collectively count for **15%** of the course grade, with all papers weighted equally. A thoughtful response earns full credit, and you may miss up to two minute papers, for any reason, without penalty. Only students attending that class meeting may submit, unless an approved accommodation provides otherwise. Minute papers cannot be made up. Beyond the two free misses, a missed paper earns a zero. The exception is an extended absence — for example, a multi-week illness supported by a dean's note — in which case the affected papers are excluded from the grade calculation.

Participation requirement. To receive course credit, you must submit a thoughtful response to at least two-thirds of the minute papers for which you are eligible. You are not counted as eligible for a paper missed because of an approved absence or accommodation. The two free misses protect your grade; they do not reduce how many papers you must submit to meet this threshold.

Bonus Questions

Each assignment includes optional extensions worth up to one additional point on the final course percentage, for a maximum of four points across the semester. Bonus work can raise your grade but cannot lower it, replace required work, or satisfy the completion requirement below.

Grading Scale

Brown awards **A**, **B**, **C**, or **NC** (no credit); there is no D or F. An NC appears on the internal academic record, but neither the course nor the NC appears on the external transcript. Grades follow a fixed scale:

Final percentage	Grade
90 and above	A
80–89	B
70–79	C
below 70	NC

The course is available on an **S/NC** basis, with an S corresponding to the C threshold (70%). All coursework remains required under either grade option.

Key Dates

Assignment deadlines are in the table above. The rest of the graded calendar:

[2] This use of low-stakes retrieval is supported by research on the testing effect; see H. L. Roediger III and J. D. Karpicke, "Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention," *Psychological Science* 17 (2006), 249–255, doi:10.1111/j.1467-9280.2006.01693.x; and C. Yang et al., "Testing (Quizzing) Boosts Classroom Learning," *Psychological Bulletin* 147 (2021), 399–435, doi:10.1037/bul0000309.

Date	What is due	Weight
Thu 10/22	**Exam 1** — in class	9%
Thu 11/19	Project teams registered on Canvas	—
Tue 11/24	**Project proposal** — one per team	3%
Tue 12/1	**Exam 2** — in class	14%
Mon 12/21	**Project report & poster session** — 9:00 AM	12%
Fridays	Research trail entries — 9/25, 10/9, 10/30, 11/13 (best three count)	7%
Most classes	Minute papers (when offered; two free misses)	15%

Assignments, trail entries, and the project proposal are due at 11:59 PM; exams are in class. Assignment 4 is due before Thanksgiving recess, and no project deliverable is due during the recess or on the first day back.

Tentative Schedule

The schedule is subject to change. Please check Canvas for the most up-to-date version.

Date	Topic / Deadline
Thu 9/10	Lecture 0 — Welcome & what is NeuroAI
Thu 9/10	*Assignment 1 released — due Tue 9/29*
Theme 1. Describing representations: spaces, similarity & geometry	
Tue 9/15	Lecture 1 — Representations, spaces & metrics: images, neurons and networks as vectors; noise and averaging; the RDM
Thu 9/17	Lecture 2 — Similarity, MDS & comparing RDMs: distance measures & the metric axioms; MDS & Shepard's law
Tue 9/22	Lecture 3 — PCA & eigendecomposition: covariance, principal components, compression
Thu 9/24	Lecture 4 — Nonlinear dimensionality reduction: manifolds; Isomap, t-SNE & UMAP
Tue 9/29	Lecture 5 — Geometry of neural representations: dimensionality & linear readouts
Tue 9/29	**Assignment 1 due**
Tue 9/29	*Assignment 2 released — due Sun 10/18*
Thu 10/1	Lecture 6 — Reading and comparing representations: shattering, cross-condition generalization, mixed selectivity, RSA & CKA
Tue 10/6	Lecture 7 — Geometry, baselines & tests: flexible and abstract codes; shuffles, cross-validation, the bootstrap, noise ceilings
Theme 2. Learning representations: from neurons to deep networks	
Thu 10/8	Lecture 8 — Learning the weights: the model neuron, training objectives, the perceptron & gradient descent

Date	Topic / Deadline
Mon 10/12	*Indigenous Peoples' Day — no University exercises*
Tue 10/13	Lecture 9 — Hidden layers & backpropagation: the chain rule, XOR & the four steps; softmax, autograd & transfer learning
Thu 10/15	Lecture 10 — Training at scale, generalization & characterizing units: minibatches, schedules & Adam; overfitting & regularization; tuning curves & Gabor fits
Sun 10/18	**Assignment 2 due**; extended from Thu 10/15
Tue 10/20	Lecture 11 — The visual system & CNNs: the ventral stream, V1 and sparse coding, convolutional networks, transfer
Thu 10/22	**Exam 1 — in-class (focuses on L01–L08 and the four steps of backpropagation from Lecture 9)**
Thu 10/22	*Assignment 3 released — due Mon 11/9*
Tue 10/27	Lecture 12 — Models of the ventral stream, and what networks compute: V1 to ResNet, attribution, feature visualization
Thu 10/29	Lecture 13 — Comparing networks with the brain: encoding models, Brain-Score and their limits
Tue 11/3	*Election Day — no class (no University exercises)*
Thu 11/5	Lecture 14 — Learning without labels: autoencoders & sparse coding; contrastive & masked self-supervised learning; the brain without labels
Mon 11/9	**Assignment 3 due**; four days later than first planned

Theme 3. Dynamics, sequences & world models

Tue 11/10	Lecture 15 — RNNs, sequences & decisions: backprop through time; evidence accumulation; LSTMs & GRUs
Tue 11/10	*Assignment 4 released — due Tue 11/24*
Thu 11/12	Lecture 16 — Neural dynamics & attractors: Hopfield networks & energy-based models
Tue 11/17	Lecture 17 — Attention & the transformer: self-attention, transformers & attention maps
Thu 11/19	Lecture 18 — LLMs & predictive coding: next-token prediction & scaling; language models and the brain
Thu 11/19	Project teams registered on Canvas
Tue 11/24	Lecture 19 — World models & frontiers of NeuroAI: world models; does better AI make better brain models?
Tue 11/24	**Assignment 4 due**
Tue 11/24	**Project proposal due at 11:59 PM** — one per team
Thu 11/26	*Thanksgiving recess — no class*
Tue 12/1	**Exam 2 — in-class (focuses on L09–L19)**

Final Project

Thu 12/3	Project work begins
Tue 12/8	Project work session — *this course meets during Reading Period*
Thu 12/10	Project work / poster prep — *last class meeting*
Mon 12/21	**Project poster session** — Brown's official final-exam slot (9:00 AM, Exam Group 12); report due

Course Policies

Accessibility and Accommodations

At Brown University, accessibility is a shared responsibility that reflects our commitment to equity and inclusion. Student Accessibility Services (SAS) provides accommodations and support for undergraduate and graduate students with disabilities. To request accommodations, contact SAS at sas@brown.edu, 401-863-9588, or visit the SAS website.

I am committed to a classroom in which students from all backgrounds are treated with respect. If a course practice creates a barrier to your participation, please tell me. For short-term academic support, undergraduates may contact a Dean of the College and graduate students may contact the Graduate School.

Religious Observance

If a religious observance conflicts with a class meeting, an assignment deadline, or an exam, let me know within the first two weeks of the semester (or as soon as you know) so we can make arrangements. You will not be penalized for observing a religious holiday.

Your Well-being

This is a demanding course, but it should not come at the cost of your health. If a difficult situation is affecting your work, contact me as early as you can. You may also reach out to Brown's Counseling and Psychological Services, Health Services, Student Support Services, or an academic dean.

Extensions, Absences, and Late Work

All coursework is required to receive course credit, subject to the miss allowances and approved exceptions stated in this syllabus. This includes four assignments, the exam requirements, all three parts of the project strand, and the participation standard under *Minute Papers and Attendance*. For an assignment, completion means a substantive, good-faith attempt at the required work; it does not require correct answers or a minimum score. A blank, placeholder-only, or otherwise nonresponsive upload does not count as completion and does not stop the late clock. If a technical problem makes a timely upload unreadable, contact me promptly and provide the file you intended to submit. Outstanding work prevents course credit even if the weighted average would otherwise pass.

Late assignments. Each late day costs **1 point off your final course grade**; any part of a day counts as one day. The penalty for an assignment is capped at that assignment's 10-point weight, but the completion requirement still applies. Unless an extension or Incomplete is approved, all assignments must be submitted by the last class meeting. This policy applies only to the four programming assignments. Exams, research-trail entries, and project deliverables cannot be submitted late without an approved extension. Minute

papers cannot be submitted late or made up; approved absences are handled as described above.

Extensions and excused absences. Any extension, and any excuse from a class, an exam, or a deadline, requires official written documentation in every case—a dean's note, a note from a physician or health service, or comparable official confirmation. This is not a judgment about whether I believe you; applying one rule to everyone is the only way to be fair to the students who do not ask. You do not need to disclose private medical details to me: an official note confirming the circumstance is enough. Extension requests must be made at least eight hours before the deadline, and no extension is in effect until it has been approved—an unanswered request is not an extension. Requests made closer to the deadline, or after it, will ordinarily be considered only in a genuine emergency, in which case contact me as soon as you are able. Late enrollment is not grounds for an extension; see *Assignments: Release and Due Dates* above. SAS accommodations and religious observances are handled under their respective policies and require no separate documentation from you.

Collaboration

Outside exams, discussion, collaboration, and research on the web are encouraged, but submitted work must be your own. Acknowledge all sources and collaborators. Unless an assignment says otherwise, you may discuss ideas and approaches but may not share code, prose, or completed answers.

Use of AI Tools

AI tools (ChatGPT, Copilot, Claude, or similar) are not permitted for assignments, exams, minute papers, or research-trail entries. Spellcheck, web search, and ordinary editor completion are allowed. AI-generated or AI-debugged code, derivations, submitted prose, and paper summaries are not.

Why this rule? AI tools are useful only if you can verify their output. To verify generated code, you must understand the Python, linear algebra, and PyTorch on which the assignments are built. Using AI on the assignments removes the practice they are meant to provide. The research trail has the same restriction because its purpose is for you to read a paper and form your own assessment.

Do not assume that unauthorized AI use will go undetected. I may ask you to explain, reproduce, debug, or modify any work you submit, and the course may use additional methods to identify AI-assisted work. Any suspected unauthorized use will be documented and referred immediately to the appropriate dean as a potential violation of Brown's Academic Code. If a violation is established, consequences can include loss of credit, No Credit in the course, suspension, dismissal, or expulsion.

AI tools are permitted for the final project. Include a short disclosure naming the tools and explaining how you used them. Final-project guidance will introduce a writer–critic workflow: one AI session produces code, a second session independently tries to break it, and you decide what to accept. You must be able to explain and defend everything you submit; the teaching staff will not debug code you cannot explain.

If you are unsure whether a use is allowed, ask before submitting. Representing AI-generated content as your own violates Brown's Academic Code even when a tool is otherwise permitted.

Academic Integrity

Follow Brown's Academic Code and acknowledge all words, ideas, code, and images you borrow. Violations may result in an NC and referral to the appropriate dean. See the undergraduate Academic Code or the graduate Academic Code.

Class Recording and Distribution of Course Materials

Slides and lecture notes are released under a Creative Commons Attribution 4.0 license (CC BY 4.0): you may share and adapt them with attribution. Figures from published papers belong to their authors and publishers and are not covered by this license. Lecture recordings and transcripts are for course use only: do not reproduce or distribute them outside the course. Students with an SAS accommodation may record for private use as specified in that accommodation.

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